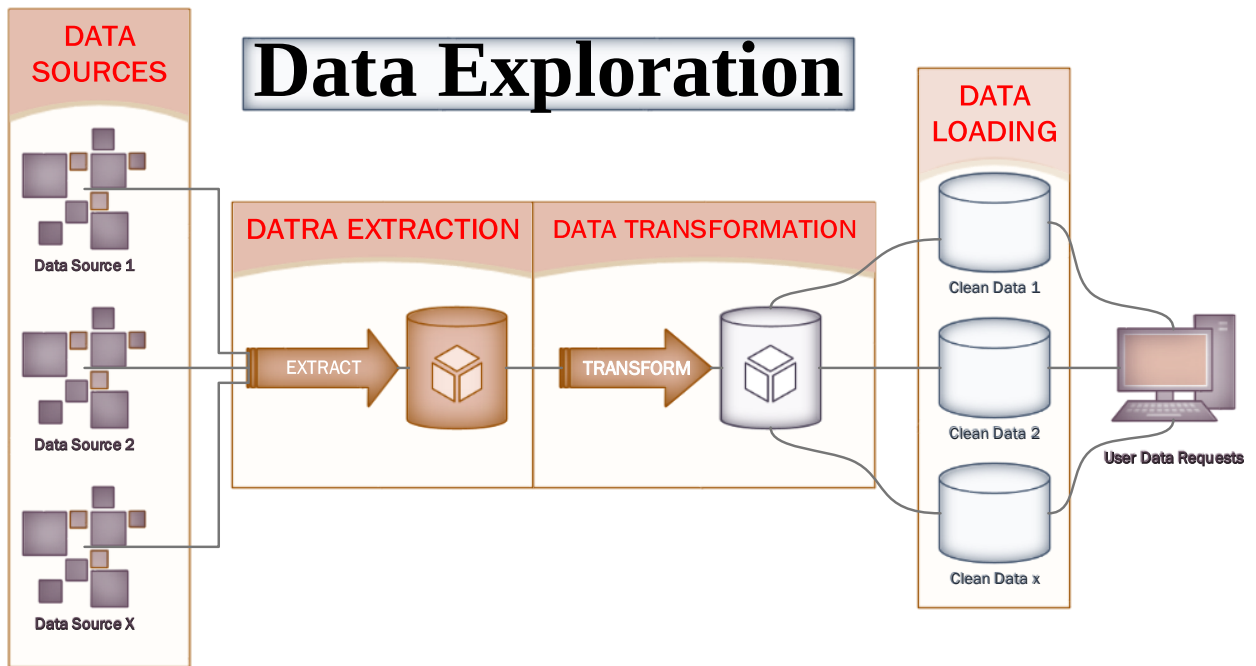




DATA EXPLORATION

The process focuses on constructing a dataset from multiple data sources to explore and model the data according to the design and database architecture. The data is being extracted from a variety of data sources to transform and load the data into its destination. The goal is to discover insights into the data and have a good understanding of any possible data quality issues. In this manner, avoid analyzing data that has not been carefully screened and reduce misleading results. The success of data analysis and reporting heavily depends on the quality of the prepared data. The aim is to describe the data by means of statistical and visualization techniques to bring important aspects of that data into focus for further analysis. Moreover, explore relationships between objects, whether there exists an association and the strength of this association. Lastly, organize and present the knowledge gained in a way that the user can use it by generating a report and, or, implementing a repeatable data mining process.

A. DATA SOURCES

Notation: Interfaces (In ←, Out →)

Existing Data Sources -> Talk to Matt for the Existing Data Sources

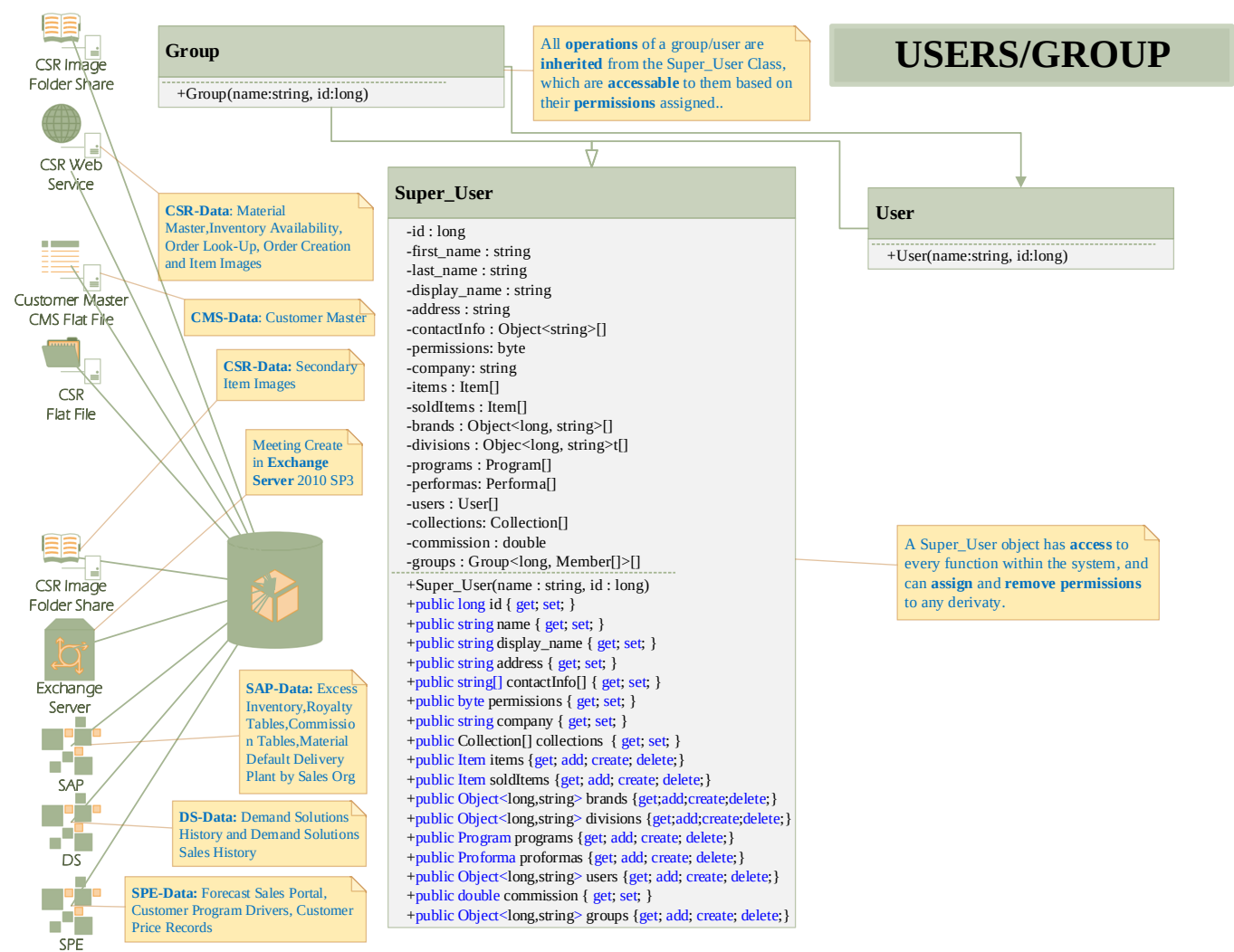
- CMS
 - o Flat File:
 - Customer Master ← Mat Tidd * Need to see the file + mean terms
- CSR
 - o Flat File:
 - Material Master ← Item/Material Record -> Matt Tidd
 - o Folder Share:
 - Item Image ← Matt Tidd (Use same code)
 - o Web Service:
 - Order Lookup ← Matt Tidd
 - Inventory Availability ←
 - Order Creation → XML-> Return

New Data Sources

- DS
 - o Flat File
 - Demand Solution History ← Forecast History
PLANNING/ Demand Solutions Data Daily Extract
 - Demand Solution Sales History ← All sales history
- CSR
 - o Flat File
 - Item Image ←
- SAP
 - o Flat File
 - Excess Inventory ← Build SAP Excess Inventory – Notify Michael
 - Finance Accrual Release B4 Documents ← LTB has to build it
 - o Flat File Export
 - Royalty Tables ← LTB has to build it
 - Commission Tables ← LTB has to build it
 - o Export
 - Material Default Delivery Plan by Sales Org → LTB has to build it
- SPE
 - o Flat File:
 - Forecast Sales Portal → Demand Solutions
 - Customer Program Drivers → ZDNN MAT
 - Customer Price Records → SAP
- OTHERS
 - o Meeting Create in Exchange Server 2010 SP3 →

Time rebate accrual injection -> Scott R.

B. DATA EXTRACTION



Demand Solution Record Attribute Fields – Record Managed Item/Plant/Customer Level

- a) **Desc.:** Material Description from SAP
- b) **UPC Code:** Code
- c) **Birthdate:** Date record was created in Demand Solution
- d) **Disc. Date:** Date Forecast is to be discontinued in Demand Solutions
- e) **Class:** SAP **Cross Plant Status** for Item (for example: 10, 21, 40, etc)
- f) **FCST:** Forecast flag that denotes whether we are allowing Demand Solution to Calculate a Forecast.
- g) **S&OP:** Flag that denotes whether we are flagged, this Item to be included in the Forecast Error KPI.
- h) **Int Ctr:** Interpolation counter that counts the number of Months in which we record Positive Sales.
- i) **Formula:** Denotes the date of latest update of the **System Forecast**, and shows the current formula being used.
- j) **Forced Formula:** Denotes if we are “hard coding” specific Forecast Formula to be used each Month.
- k) **MRP:** **MRP controller** denotes which specific Demand Planner is assigned to that(a) record.
- l) **X Plant Status:** Shows SAP **Company Wide Status**
- m) **Plant Specific Status:** Shows SAP SKU Status at Plant Level (i.e. 1000 v SYML)

Item Status Attributes Fields – We will need a few more though.

CA = Coming Attraction

RTG = Ready to Go

AIS = Always in Stock

DNR = Do not Reorder

DISC = Discontinued

Material Record Attribute Fields – Record Managed Item/Plant/Customer Level

ITEM

ITEM_DESC

BRAND_DESC

CATEGORY_DESC

VARIANT_DESC"

PATTERN_DESC

BASIC_MATERIAL

XSTATUS

LENGTH

WIDTH

HEIGHT

DIMENSION_UNIT

WEIGHT

WEIGHT_UNIT

PLANT

UPC

COLOR

DIST_CHANNELS

Demand Solution Data Daily Export Attribute Fields –

Company: Company
Division: Division
Ship To: Customer

Item: Item
Future Price: Price

Adjusted History 1:
Adjusted sales history (First record date - 1 month)
Birth Date: Record Start

Adjusted History 12:
Adjusted sales history (First record date - 12 month)
Discontinue Date: Record Disc

Demand Solution Data Daily Import Attribute Fields –

Company: Company
Planner: Planner
Available User Field 3: MRP Controller
Division: Plant
Ship To: Customer
Original History 1: Sales history (First record date - 1 month)
Original History 13: Sales history (First record date - 13 month)
Original History 25: Sales history (First record date - 25 month)
Adjusted History 1:
Adjusted sales history (First record date - 1 month)
Birth Date: Record Start

Item: Item
SOP Flag: SOP Flag
Forecast Flag: Forecast Flag
Future Price: Price
Unsigned Error: Absolute Error
Original History 12: Sales history (First record date – 12 month)
Original History 24: Sales history (First record date - 24 month)
Original History 36: Sales history (First record date - 36 month)
Adjusted History 12:
Adjusted sales history (First record date - 12 month)
Discontinue Date: Record Disc

Demand Solution Data Extract Attribute Fields –

Company: Company
Planner: Planner
Available User Field 3: MRP Controller
Division: Plant
Ship To: Customer
Original History 1: Sales history (First record date - 1 month)
Original History 13: Sales history (First record date - 13 month)
Original History 25: Sales history (First record date - 25 month)
Adjusted History 1:
Adjusted sales history (First record date - 1 month)
Adjusted History 13:
Adjusted sales history (First record date - 13 month)
Adjusted History 25:
Adjusted sales history (First record date - 25 month)
Adjusted Forecast 1: Forecast (First record date + 1 month)
System Forecast 1 :
System forecast (if applicable) (First record date + 1 month)
Birth Date: Record Start
Historical Price1: Price (First record date - 1)
Historical Price13: Price (First record date - 13)
Interpolation Counter: IntCounter

Item: Item
SOP Flag: SOP Flag
Forecast Flag: Forecast Flag
Future Price: Price
Unsigned Error: Absolute Error
Original History 12: Sales history (First record date – 12 month)
Original History 24: Sales history (First record date - 24 month)
Original History 36: Sales history (First record date - 36 month)
Adjusted History 12:
Adjusted sales history (First record date - 12 month)
Adjusted History 24:
Adjusted sales history (First record date - 24 month)
Adjusted History 36:
Adjusted sales history (First record date - 36 month)
Adjusted Forecast 12: Forecast (First record date + 12 month)
System Forecast 12:
System forecast (if applicable) (First record date + 12 month)
Discontinue Date: Record Disc
Historical Price12: Price (First record date - 12)
Historical Price24: Price (First record date - 24)

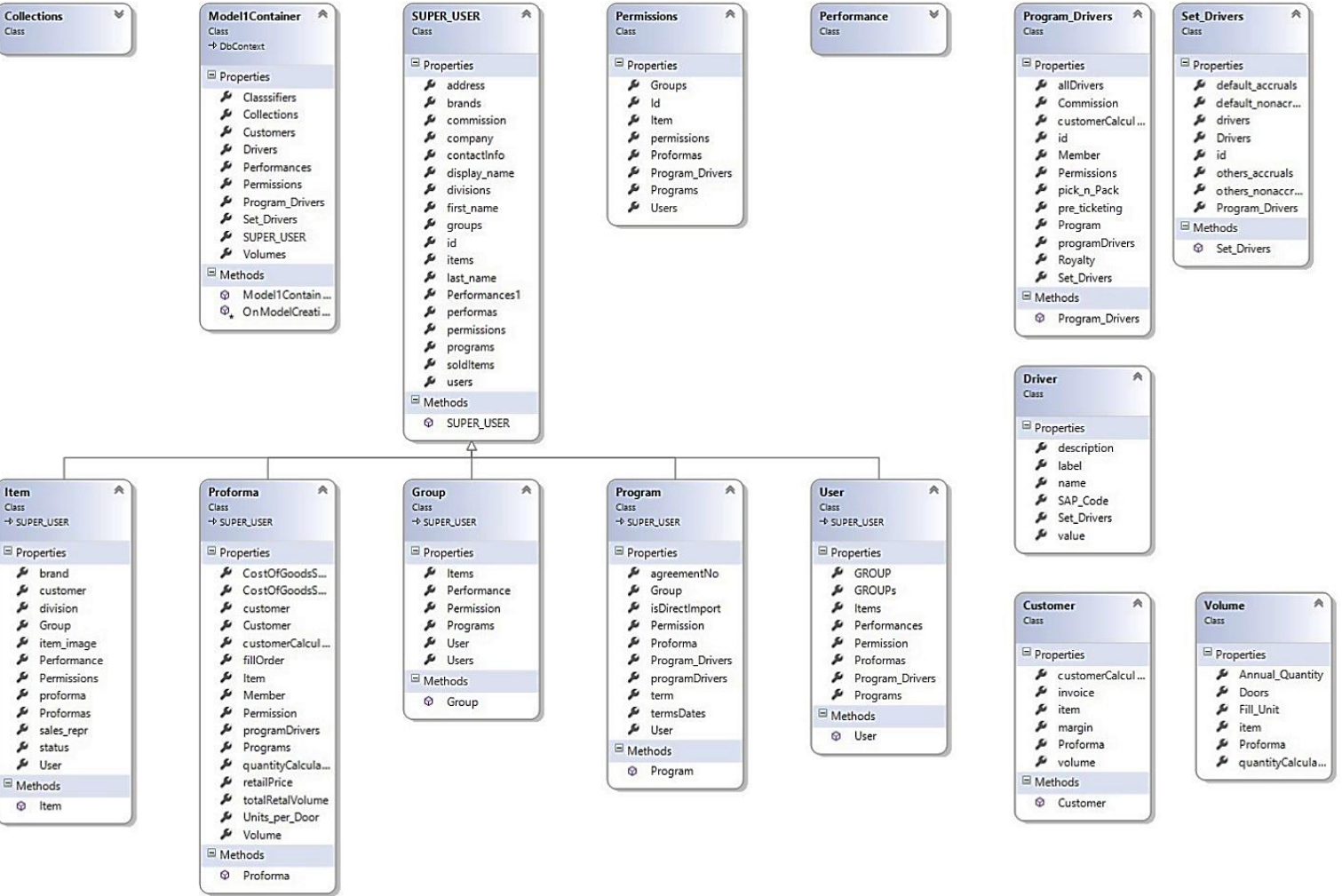
Planning Attribute Fields

Field Label	Source	Record Start	Demand Solutions, default to First time record is created though.
Company	Admin Table		
Planner	Demand Solutions	Record Disc	Come from user input
Customer	Use existing table from CMS system	# of doors	Will come from the proforma create in this portal or user input
Division	Use existing table from CMS system	Forecast Flag	Demand Solutions
Item	CSR Material Master	S&OP	Demand Solutions
Plant	SAP Material Table	IntCounter	Demand Solutions
PY Forecast	Initial Demand soluutuons, then save forecast end of each month	MRP	Demand Solutions
PY Sales	Initial Demand soluutuons, then save forecast end of each month	CO (ABC for Company)	Demand Solutions
Var	Calculation	PL (ABC for planner)	
Forecast	Demand Solutions, or user input	CU (ABC for Customer)	
MTD Sales	If possible use the web service, else will supply daily import file.	Err (XYZ)	CSR Material Master (new field needs to be added)
Open Orders	If possible use the web service, else will supply daily import file.	Net Error	Calculation using last 12 months of past data
UPC	CSR Material Master	Abs Error	Calculation using last 12 months of past data
X-Plant-Status	CSR Material Master	Lag Error	Calculation using last 12 months of past data, use saved forecast offest from items leadtime in months.
Item Status	Not sure yet.	Lead-time	CSR Material Master (new field needs to be added)
Proforma Qty	Will come from the proforma create in this portal.	Excess:	Daily import from flat file.

Planning Attribute Fields –

Field Label	Source Formula	Example	Internal notes
Company	Admin Table	Two fields concatenate, Code - Desc	SYUS - US
Planner	Demand Solutions	Planner name in DS needs to get mapped to Active directoy User	Scott Wit
Customer	Use existing table from CMS system	Customer Desc - Customer Code	Bed Bath & Beyond
Division	Use existing table from CMS system	Division Desc - Division Code - User cannot edit data, its based on the Material	
Item	CSR Material Master	Material - Material Desc	
Plant	SAP Material Table	Default to default plant on material master load.	
PY Forecast	Initial Demand solutuons, then save forecast end of each month		
PY Sales	Initial Demand solutuons, then save forecast end of each month		
Var Calculation	PY Sales - PY Forecast ((PY Sales - PY Forecast)/PY Sales)		
Forecast	Demand Solutions, or user input	Need a checkout and in process.	When checkout is run it exports the forecast data to a flat file. During checkout the forecast on existing records cannot be changed, new records can be added and those can have forecast changes. For checking-in a flat file will be dropped into a shared location and it will be imported, if successful the forecast fields are unlocked for editing. if there is excess qty on the record all fields are white, if the excess is consumed or does not exist turn the cells in lead-time red.
MTD Sales	If possible use the web service, else will supply daily import file.		
Open Orders	If possible use the web service, else will supply daily import file.		
UPC	CSR Material Master		
X-Plant-Status	CSR Material Master		
Item Status	Not sure yet.		
Proforma Qty	Will come from the proforma create in this portal.		
Record Start	Demand Solutions, default to First time record is created though.		
Record Disc	Come from user input		
# of doors	Will come from the proforma create in this portal or user input		
Forecast Flag	Demand Solutions		
S&OP	Demand Solutions		
IntCounter	Demand Solutions		
MRP	Demand Solutions		
CO (ABC for Company)	Demand Solutions		
PL (ABC for planner)			
CU (ABC for Customer)			
Err (XYZ)	CSR Material Master (new field needs to be added)		
Net Error	Calculation using last 12 months of past data(Sum(Forecast) - sum(Sales))/ sum(forecast)		
Abs Error	Calculation using last 12 months of past dataabs(Sum(Forecast) - sum(Sales))/sum(forecast)		
Lag Error	Calculation using last 12 months of past data, use saved forecast offest from items leadtime in months. abs(Sum(Forecast) - sum(Sales))/sum(forecast)		
Lead-time	CSR Material Master (new field needs to be added)	Planned delivery time + GR processing time convereted to calendar days, if no fixed vendor use 120 days	
Excess:	Daily import from flat file.		

C. DATA TRANSFORMATION



D. DATA LAODING

